WATERLOO ROCKETRY

Next-Gen Ground Support Software System (Orion)

Preliminary Design Review

Qian Qian
Software Lead

Shrey Shingala
Software Lead

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SECTION 01

Overview, Motivation & Feasibility

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Overview

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Motivation & Feasibility

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SECTION 02

Requirements

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Requirements Overview

The GitHub issues `2026-2027-software-issues#51` contains and is the root parent of issues containing:

- general technical directions
- design (tasks and decisions)
  - specification
  - API Protocol and Data Model
- implementation (requirements and tasks)

Please review its content.

General Requirements

Requirement	Motivation
Omnibus MUST remain production-ready until Orion is fully production-ready and is fully replacing Omnibus.	Allow smooth transition and ensure fallback option exists while issues are ironed out.
Orion SHOULD provide one authoritative API server for telemetry, shared state and command routing.	Provides a stable boundary between sources and consumers, and avoid system-wide breaks when a source or consumer is updated.
Public interfaces SHOULD be typed, versioned, documented and machine-checked.	Standardized interfaces to ensure safe interaction and validation of data.
Complete and well organized documentation MUST be provided for all public interfaces, including API, data model, and configuration.	Ensure that the system is usable and mainytainable by future developers and users.
The deployment of the system MUST be reproducible.	Ensure that the system can be deployed in a consistent manner.
The system MUST be easy to configure and the configuration methods MUST be documented.	Ensure that the system can be configured by all users. (Abandoning the current Python-script-based configuration system.)

SECTION 03

Explanation of Key Technological Choices

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